# Limits and Continuity

1. Evaluate the limit:

$$\lim_{x\to-\infty} \frac{x^2-\sin(x)}{\sqrt{x^4+1}}$$

2. Determine the value of $a$ and $b$ so that the function f is continuous everywhere.

$$f(x)=\begin{cases} ax-3, & x<1, \\ \dfrac{bx-1}{2}, & 1\le x\le 3, \\ \dfrac{x^2-(b-2)x+a-4}{x-2}, & x>3. \end{cases}$$

3. Evaluate the limit:

$$\lim_{x\to5^-} \frac{|x-5|}{2x^2-11x+5}$$

4. Prove that

$$\lim_{x\to-2}(4x^2+7x+1)=3$$

5. Determine the horizontal asymptote of the function:

$$f(x)=\frac{x^4-2\cos(2x)}{2\sqrt{x^8-2}+1}$$

6. Evaluate the limit:

$$\lim_{x\to2} \frac{x^3+x^2-4x-4}{x-2}$$

7. Let $f(x)=x^3-x^2+2x-5$. Use the Intermediate Value Theorem to show that there exists $c\in(0,2)$ such that $f(c)=-3$ and find the value of $c$.

8. Suppose $f(x) = x^2 - 2x + 4$. Show that $f$ takes an extreme value on $[0, 3]$ and determine its extreme values.

9. Evaluate the limit:

$$\lim_{x \to 0} \frac{e^{4x} - e^{3x}}{5x}$$

10. Evaluate the limit:

$$\lim_{x \to 5^+} \left( \frac{1}{x-5} \right) \left( \frac{x-1}{4} - 1 \right)$$

11. Determine the horizontal asymptote of the function:

$f(x) = \dfrac{2[\![x]\!] + 1}{x^2}$. Where $[\![x]\!]$ denotes the greatest integer $x$

12. Evaluate the limit:

$$\lim_{x \to 2} \frac{\sin(x^2 - 4)}{5x - 10}$$

13. Determine the value(s) of $x$ that make the function $f$ discontinuous and identify the type of discontinuity:

$$f(x) = \begin{cases} 3x + 4, & x \leq -1, \\ \dfrac{x-3}{x^2 - x - 6}, & -1 < x \leq 4 \\ \dfrac{x-5}{x^2 - 3x - 10}, & x > 4. \end{cases}$$

14. Evaluate the limit:

$$\lim_{x \to 6} \frac{\sqrt[3]{x-5} - 1}{x - 6}$$